Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering

B.Tech. II (CSE) Semester-III
Database Management Systems (CS203)

Academic Year: 2026-27

Practical Assignment – 10
Date: 12th October 2026

PL/SQL Assignment: To develop Backend Logic for your registered project

For your registered project, which allows the users to interact with a frontend UI. To design **CRUD (Create, Read, Update, Delete) operations**, **cursors**, **procedures, functions** and **triggers**, implement the backend components that handle data integrity, business logic, and transaction management using PL/SQL for this UI.

Write appropriate PL/SQL stored program to perform the following:

1. Create individual stored procedures to handle backend data manipulation:
   A. Insertion Procedure
   B. Update Procedure
   C. Deletion Procedure
2. Create different user's accounts with their roles like naïve, application developer and DBA.
3. Perform 1, to manage User Login data.
4. Write a stored procedure using an explicit cursor to iterate through your table data.
   A. Without limit
   B. With usage of limit
5. Create function for paginated search for your table data with and without filtering conditions.
6. Write a function to return the sum of any column with numeric data.
   A. Without grouping
   B. With grouping
7. Create a BEFORE trigger that automatically checks any two data integrity constraint on your table data.
8. Create a statement trigger to convert some char/varchar data to Uppercase letters.

9. Create an AFTER trigger that automatically logs any data modification (INSERT, UPDATE, or DELETE) performed on your table data into the application_audit_log table.
10. Create a trigger to prevent any modifications to any one of your project tables, if accessed after the standard office hours working days (for your implementation, you can consider 2 to 4 pm and Monday) with the appropriate message.

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